

Kurve RSC

Learning Relational Representations with the Downstream Predictor in the Loop

Kurve AI

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1 Abstract

Kurve RSC is a validation-guided system for learning how a relational database should be compressed into a model-ready table. It does not assume that one fixed flattening of a database is best. Instead, it proposes alternative relational programs—different graph depths, feature-family stages, and annotation policies—executes each program, fits a downstream learner on the resulting frame, and uses held-out predictive performance to decide which representation should survive.

This makes feature construction and model fitting one integrated learning process. The relational configuration changes the shape and meaning of the training frame; the learner reveals whether that representation carries useful signal for the task. In the current implementation, CatBoost is the bundled downstream learner and classification and regression candidates are selected by validation AUROC and MAE, respectively. The dataframe boundary is deliberately learner-neutral. An experimental TabPFN v3 adapter demonstrates that boundary, while CatBoost remains the reference production backend.

Kurve RSC is built with [GraphReduce](#), the open-source engine that executes relational compute graphs. Kurve RSC is the larger learning and model-lifecycle system around that engine: it defines the candidate space, binds tasks and cutoffs, compares the resulting representations through a downstream learner, selects a configuration, freezes the relational execution plan and feature schema, and produces a fitted artifact for validation and inference.

This white paper focuses on that optimization idea. The companion [KurveRSC technical report](#) contains the detailed treatment of cutoff semantics, relational reduction, feature families, system budgets, benchmark protocol,

and reproducibility.

2 The central idea

Most relational machine-learning pipelines make two decisions in sequence: first, an engineer fixes a flattening strategy; then, a model is optimized on the resulting table. Kurve RSC moves the first decision inside the validation loop.

Let a relational configuration be

$$q = (d, \mathcal{A}, a, b),$$

where d is graph depth, $\mathcal{A}$ is the cumulative set of enabled feature families, a is the automatic-annotation policy, and b denotes feature budgets. For a database $\mathcal{D}$, task rows L , and cutoff c , GraphReduce executes the relational program selected by q and Kurve RSC materializes the entity-grain frame

$$F_q(c) = \text{Materialize}(\mathcal{D}, L, c; q).$$

Kurve RSC then fits a downstream learner M_q on the training portion of that frame and evaluates it on the validation portion:

$$M_q = \text{Fit}(F_q^{\text{train}}), \quad J(q) = \text{Metric}(M_q, F_q^{\text{validation}}).$$

The winning relational representation is therefore the one supported by the performance of a fitted model, not the one that merely creates the most features:

$$q^* = \begin{cases} \arg \max_q J(q), & \text{classification AUROC,} \\ \arg \min_q J(q), & \text{regression MAE.} \end{cases}$$

This is joint in the systems sense: representation construction and learner evaluation form one objective loop. It is not an end-to-end differentiable graph in which gradients pass through SQL. Kurve RSC performs deterministic, validation-guided configuration search over complete feature-and-model trials.

3 Why frame shape is part of learning

Every candidate is a hypothesis about which relational signal should reach the prediction grain.

A shallow, **base**-only candidate may reduce only nearby tables and emit a narrow frame containing stable schema-aware rollups. Increasing depth admits longer foreign-key paths, so additional tables can contribute summaries to the root entity. Adding **temporal** introduces windowed numeric behavior. Adding **sequence** preserves rates, recent activity shares, burst ratios, and active spans. Adding **conditional** introduces bounded, condition-specific composition, while **episode** counts exposure and distinct events. Switching automatic annotations changes the predicates and gated statistics available to those reductions.

For candidate q , write the model-facing frame shape as

$$F_q \in \mathbb{R}^{n_q \times p_q}.$$

The row count n_q is chiefly determined by the eligible labeled entities and cutoffs. The feature count p_q and the semantic lineage of those columns are functions of depth, reachable tables, family selection, annotation policy, source types, and available values. Graph execution or label attachment can also make row counts differ in practice, so Kurve RSC records both dimensions for every completed trial.

The important point is that these are not cosmetic transformations of one fixed matrix. A depth-1 **base** frame and a depth-2 **base + temporal + sequence + conditional** frame are different representations of the task. Wider is not automatically better: extra columns can be redundant, sparse, expensive, or easier for the learner to overfit. Validation performance is the feedback that resolves that ambiguity.

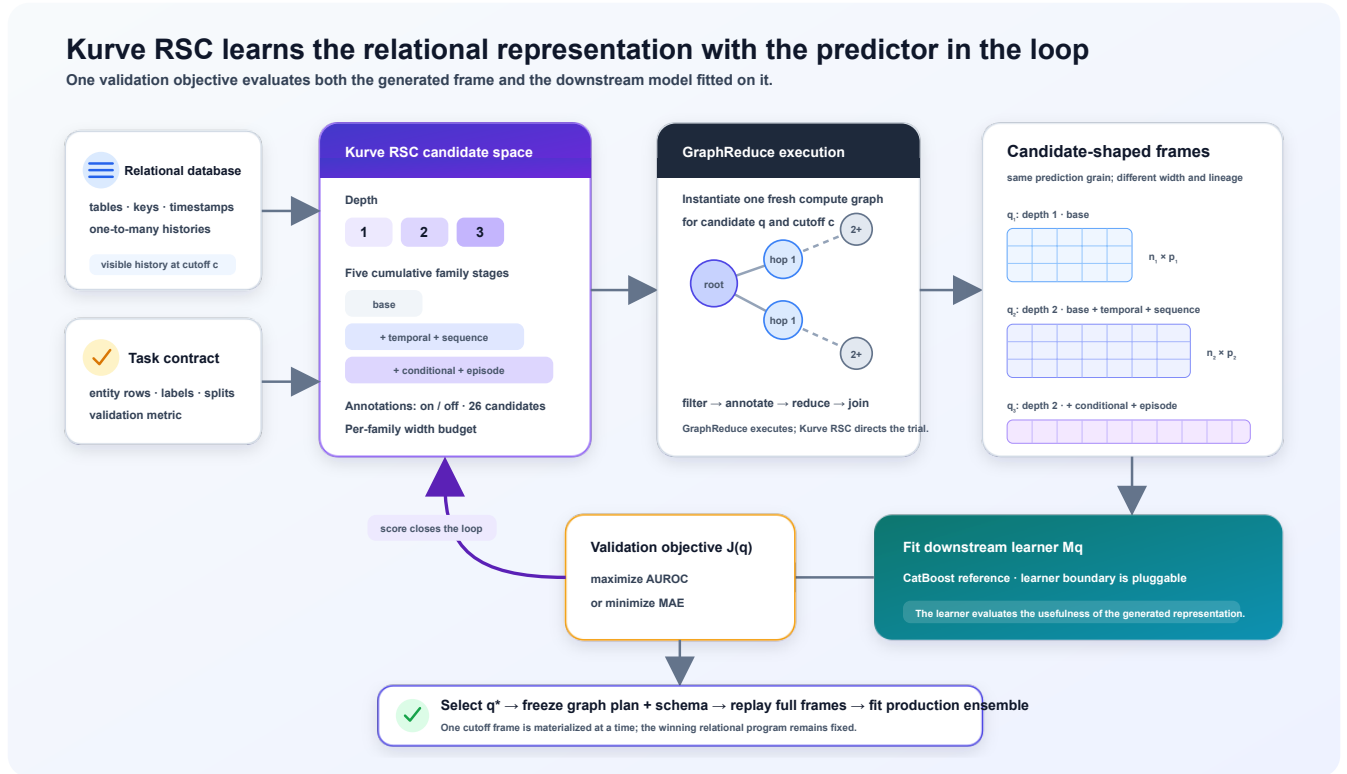


Figure 1: Kurve RSC jointly evaluates relational configuration and downstream model performance. Each candidate changes graph reach and feature policy, producing a differently shaped frame before learner fitting and validation scoring.

The current default candidate schedule is deterministic and beam-pruned:

Search axis	Current default
Relational depth	1, 2, and 3 hops back toward related tables
Feature-family subset	base plus independent subsets of temporal , sequence , conditional , and episode
Automatic annotations	enabled and disabled
Source-column budgets	utility-ranked 4 columns per family by default; optional wider budgets
Per-source / propagated expansion caps	32 derived features / 1 canonical continuation
Downstream learner	CatBoost
Classification objective	maximize validation AUROC
Regression objective	minimize validation MAE

Subsets containing only base, temporal, and sequence are eligible for depth 3. Subsets containing conditional or episode are limited to depths 1 and 2. One column budget has 72 potential shapes, but the adaptive forward beam normally materializes no more than 24. Optional wider source-column budgets add another candidate lattice and promote only a bounded set from the narrower budget. Users can supply explicit **GraphConfig** candidates when this space is not appropriate.

4 The coarse-to-full learning protocol

Kurve RSC separates economical configuration discovery from the production fit. This distinction matters because relational materialization may be more expensive than fitting the downstream model itself.

The reference RelArena profile uses a three-fidelity funnel:

```

candidate lattice
  |
  v
10K-row screening
  | confirmation_top_k=8
  v
up to 8 diverse GraphConfigs on 50K rows
  | rerank_top_k=3
  v
top 3 on three sequential full-data cutoff folds
  |
  v
raw stability-adjusted winner

```

`confirmation_top_k` is the maximum number of distinct relational configurations promoted from inexpensive screening to medium-fidelity confirmation. It is not a CatBoost hyperparameter, a frame count, or a cutoff count. The selector first retains the raw objective leader and the complexity-aware screening recommendation. It then seeks score-ranked representatives of the available optional feature families, a deeper graph, and the opposite automatic-annotation policy before filling any remaining slots by raw score. Duplicate shapes are removed, so fewer than eight may be confirmed when the successful search frontier is small.

The medium-fidelity row count is controlled by `sample_rows`; the RelArena profile sets it to 50,000. When `search_full_data=True`, every beam-admitted candidate is evaluated directly on complete source tables, the separate confirmation stage is skipped, and `confirmation_top_k` has no effect.

4.1 Stage 1: search on a controlled point-in-time frame

Each candidate receives a fresh GraphReduce graph because graph execution is stateful. The current implementation first caps each search source at `screening_rows` and searches on the latest eligible training cutoff. The surviving diverse candidates are then rebuilt at the `sample_rows` confirmation fidelity. For every evaluated candidate, Kurve RSC:

1. builds the depth-bounded relational graph and applies the candidate’s feature policy;
2. learns a feature-operation plan from training data;
3. materializes aligned training and validation frames under that plan;
4. normalizes model inputs while freezing the training-derived column order;
5. fits CatBoost and records validation score and its standard error, feature count, row counts, feature time, and model time; and
6. preserves failures and resource-related skips in the trial audit trail.

The configuration with the best objective score becomes `best_config`. Kurve RSC also exposes `recommended_config`: the simplest candidate within a small performance tolerance of the best. The fixed 0.002 AUROC and 0.5% MAE floors are combined with estimated validation uncertainty, so extra complexity must produce a statistically meaningful gain. This makes representational cost visible without replacing the raw metric winner.

4.2 Stage 2: select and freeze one relational representation

Each configuration is evaluated on one latest eligible training frame and the declared validation data. With full-data search enabled, that frame contains all eligible rows rather than a sample. The best raw metric and the complexity-aware recommendation are retained in the audit trail. The strongest three raw medium-fidelity candidates are then rescored over sequential full-data cutoff folds. Their mean validation objective is penalized by temporal score instability. The raw maximizer of this stability-adjusted objective is selected; the earlier complexity recommendation remains auditable but cannot override the full-frame evidence.

Kurve RSC freezes the selected GraphReduce execution plan and ordered feature schema. Subsequent temporal frames replay this learned representation rather than rediscovering feature operations.

4.3 Stage 3: fit the production frame ensemble

When enabled, GraphReduce’s event-cadence inference lets dated nodes and relationships replace the initial lookback grid with compact periods inferred from their observed event frequency and the configured compute horizon. Kurve RSC records those inferred periods as part of the selected plan, then restores rather than rediscovers them throughout historical replay, outer refitting, and prediction.

For timestamped problems, the production refit can use several point-in-time training frames. If the selected cutoffs are $c_1, \dots, c_K$, each compatible frame is consumed and released before the next is constructed. An independent CatBoost model is fitted per frame, and their predictions are averaged:

$$M_k = \text{CatBoostFit}(F_{q^*}(c_k)), \quad \hat{f}(x) = \frac{1}{K} \sum_{k=1}^K M_k(x).$$

Here K increases temporal coverage and usually frame height, while d and $\mathcal{A}$ control relational reach and feature width. The operation plan is learned once on the latest full-training frame and replayed at the other cutoffs, preventing each historical view from silently inventing a different feature program.

Validation is scored using only the ensemble of training-frame models. Kurve RSC then fits independent validation-frame models and adds them to the final train-plus-validation ensemble. Test or future frames replay the frozen graph plan and training-only schema with `GraphReduce(train=False)`. Feature discovery does not run again at inference time.

5 The downstream learner is a replaceable participant

Kurve RSC’s stable interface to learning is an aligned dataframe, target, split contract, and metric. This is intentionally narrower than a model framework and broader than a CatBoost-specific feature generator.

CatBoost is the primary backend today because it is a strong conventional learner for mixed numerical and categorical data and can expose the value of a relational representation without requiring a bespoke neural architecture. The implementation currently holds the supplied CatBoost parameter policy fixed while it searches the relational configuration. Each independent production-frame model receives the declared tree budget.

TabPFN is a natural alternate backend. From Kurve RSC’s perspective, the relational system compiles database history into the tabular context on which a foundation model can condition or be fine-tuned. Context length, row budget, and feature budget will affect the feasible candidate space, but they do not change the contract: a candidate is judged after the chosen learner consumes its frame. The same separation can support other gradient-boosted trees, linear models, neural tabular models, or AutoML systems as adapters are added.

Learner neutrality should not be confused with implemented parity. CatBoost is the packaged reference path and the only backend currently supporting multi-frame production fitting. The experimental TabPFN v3 adapter supports single-frame fitting. Kurve RSC’s architectural contribution is that additional learners do not require redefining point-in-time relational construction.

6 Kurve RSC and GraphReduce: a deliberate boundary

Kurve RSC and GraphReduce are related projects with distinct responsibilities. GraphReduce deserves direct credit as the relational execution engine on which the current Kurve RSC implementation is built. Kurve RSC should nevertheless be understood as its own learning system, not as a new name for GraphReduce.

Kurve RSC owns the learning system	GraphReduce supplies the execution substrate
Task and schema adaptation	Table-node and relationship-edge abstractions
Candidate-space construction	Ordered relational compute operations
Depth, family, annotation, and budget search	SQL generation and execution
Cutoff scheduling and entity-frame attachment	Temporal filtering primitives
Sampled search and full-data winner rebuild	Many-to-one reduction and feature propagation
Validation objective and complexity recommendation	Reusable feature-family implementation
Frozen-plan/schema lifecycle and prediction API	Compute-graph runtime machinery

Kurve RSC owns the learning system	GraphReduce supplies the execution substrate
Downstream learner integration and fitted artifact	Execution of the relational program requested by Kurve RSC

A useful analogy is that GraphReduce is the compiler/runtime for a relational program, while Kurve RSC decides which programs to propose, runs the empirical selection process, and binds the winning program to a predictive model and its lifecycle. The value of Kurve RSC lies in this orchestration and learning loop: it converts graph configuration from an engineering constant into a validation-tested hypothesis.

The boundary also makes both projects more useful. GraphReduce remains a general-purpose relational feature-engineering project that can be used without Kurve RSC. Kurve RSC can evolve its search policy, task adapters, model backends, and experiment semantics while continuing to rely on a specialized relational engine.

7 Evaluation context and attribution

This white paper intentionally contains no performance table. The detailed benchmark report presents Kurve RSC classification and regression results, run disclosures, ablations, and caveats. Readers should consult that report before drawing empirical conclusions.

For classification context, the external comparison columns reproduced in the technical report originate from **Appendix D, Table 4 of the August 2026 Prior Labs RelArena- α report** [6]. Credit for that table, its RelArena evaluation work, and the reported TabPFN-Rel, RDBLearn, RelGNN, and RelGT results belongs to Hayler, Flöge, Arazi, Ranjan, Leskovec, Purucker, Hutter, Hollmann, and the Prior Labs Team. Those external values were not produced by the Kurve RSC repository.

RelArena comparisons are valuable context, but they are not automatically a controlled within-harness experiment. Package revisions, database state, tuning budget, cutoff treatment, and execution protocol can differ. Kurve RSC therefore treats validation discipline and full run disclosure as part of the result, not as footnotes to it.

RelBench supplies the official relational datasets, task definitions, temporal splits, and evaluators used by these experiments [3]. Kurve RSC does not redefine those benchmark labels or metrics; it learns a representation and predictor for the task contract RelBench provides.

8 What Kurve RSC is optimizing—and what it is not

Kurve RSC optimizes the predictive utility of a relational representation under a declared search space and validation split. In the current system, that includes depth, independent feature-family subsets, automatic annotations, and a utility-ranked source-column budget funnel. The objective reflects the combination of the generated frame and fitted learner.

It does not yet claim to solve every related optimization problem:

- Search is finite and staged, not continuous or gradient-based.
- The default search does not jointly sweep arbitrary CatBoost hyperparameters or learner families; caller-supplied model parameters are held fixed across relational candidates.
- Screening uses one latest training frame, so a weak one-frame rank can still exclude a configuration whose value appears only with additional temporal coverage.
- The primary selection objective is predictive performance. Runtime and feature count inform the simpler recommendation but are not yet a general multi-objective cost optimizer.
- Validation can select among proposed configurations only; it cannot recover useful signal excluded from the candidate space or unavailable under the declared point-in-time history.

These constraints are purposeful. They make each trial auditable and keep test labels outside configuration choice. They also identify a clear roadmap: adaptive candidate generation, cost-aware multi-objective search, learner-aware feature budgets, multiple downstream backends, and policies that decide how many full-training frames are worth materializing.

9 Conclusion

Kurve RSC learns more than model parameters. It learns which relational representation should be presented to the model.

Each candidate configuration reaches a different distance into the database, preserves a different family of signals, and produces a frame with its own shape and lineage. A downstream learner is fitted on that frame, and held-out performance closes the loop. The selected representation is then rebuilt on full data, frozen as an execution plan and schema, and carried into a production fitted artifact.

GraphReduce is the enabling relational engine and an important open-source foundation for this work. Kurve RSC is the independent system that turns that engine into a validation-guided representation-learning process coupled to an interchangeable downstream learner. CatBoost is the practical starting point; TabPFN and other learners extend the same idea. The enduring abstraction is the loop itself: propose a relational view, learn on it, validate it, and keep the view whose usefulness survives contact with the prediction task.

10 References

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